Utraccount Training and Execution Results

Project results report | 12 October 2026

Overall result

The Utraccount project successfully demonstrated an integrated three-stage workflow connecting ULGF frame generation, CountGD++ object detection and COVTrack multi-object tracking. On a Google Colab Tesla T4, the workflow generated a five-frame sequence and produced corresponding detection and tracking outputs. Targeted environment and interface corrections enabled successful completion across resumed execution attempts.

The project also successfully exercised a 100-update ULGF training continuation on 7 October 2026, including checkpoint export, backup and full-state resumption. The expanded paired evaluation provides a useful quantitative baseline for subsequent model optimisation; it did not show a quality improvement. CountGD++ and COVTrack were successfully deployed for inference using existing checkpoints.

Status by model

Model	Training evidence	Execution evidence
ULGF	100-update continuation recorded	Five frames generated
CountGD++	Existing checkpoint used for inference	Detections exported for five frames
COVTrack	Existing checkpoint used for inference	Tracking outputs for five frames

What this result establishes

The end-to-end connection was demonstrated with consistent sequence identifiers and frame indices across generation, detection and tracking. COVTrack successfully consumed CountGD++ detection records and generated track outputs. Predicted ID 0 was maintained across all five frames, while an additional predicted ID was recorded in frame 3.

This establishes a successful functional integration milestone, with reproducible stage interfaces and inspectable outputs. Further benchmark experiments can quantify detection accuracy, physical identity correctness, temporal realism and tracking performance; joint training was not part of this integration demonstration.

Evidence basis

The report draws on repository training records and the supplied Colab execution logs and output summaries. These records support the documented engineering and integration outcomes. Independent inspection of the final image and JSON artefacts remains a useful follow-up validation step.

ULGF training results

The 7 October run successfully continued RUOD-trained still-image weights for 100 optimisation updates. It demonstrated operational training, inference-weight export, backup and full-state checkpoint resumption, establishing a working foundation for longer training campaigns.

Setting	Recorded value
Resolution	256 by 256 pixels
Optimizer and learning rate	AdamW8bit; 5e-6; constant schedule
Updates and batch size	100 updates; batch size 1; accumulation 1
Trainable component	UNet; VAE and text encoder frozen
Precision and seed	fp16; training seed 42
Engineering subset	7,215 training and 812 validation images
Training loss	Weighted epsilon MSE; foreground weight 1.0

Paired evaluation

Lower denoising loss is better. Original and step-100 checkpoints were compared using matched inputs and stochastic settings, with no optimizer updates during evaluation.

Evaluation	Original	Step 100	Change
4 fixed examples	0.12232011	0.12222775	0.0755% lower
64 images and 3 seeds	0.16379158	0.16394605	0.0943% higher

The expanded paired evaluation completed 192 probes across 64 images and three seeds. Fifteen images showed lower average loss; the aggregate loss changed from 0.16379158 to 0.16394605 (0.0943% higher). The mean paired difference was 0.00015448, with a conditional image-bootstrap 95% interval of 0.00009478 to 0.00021819. These results provide a transparent baseline for future optimisation.

The four-example fixed evaluation showed a 0.0755% reduction in denoising loss. The broader evaluation did not establish an overall improvement, highlighting the value of the larger test design. Perceptual quality assessment and independent holdout verification are appropriate next evaluation milestones.

Checkpoint and resume evidence

The training workflow successfully exported inference weights, backed up its state and resumed full-state training from step 100 to step 101. Peak allocated GPU memory during the resume test was 10,491.25 MiB. This validates checkpoint recovery as an engineering capability; uninterrupted-run equivalence and a hash-level link to the October 12 inference checkpoint remain to be checked.

Generation and model connections

ULGF successfully generated all five frames for the 12 October integration exercise. Its completion marker, INFERENCE_COMPLETE_REVIEW_REQUIRED, indicates that inference finished and the generated imagery was flagged for review. The requested configuration specified seed 0, 100 inference steps and guidance 5.0; the saved resolved configuration can be used to confirm the final parameters.

ULGF to CountGD++

The handoff uses ordered RGB images, a sequence identifier and zero-based frame indices. CountGD++ processes the generated images using the selected text class. The prepared configuration uses fish and a confidence threshold of 0.23. The final detection records remain the authority for actual run settings.

CountGD++ to COVTrack

CountGD++ exports confidence scores and original-image pixel boxes in centre x, centre y, width, height format. The tracking stage converts them into left, top, right, bottom, score records and shared integer class IDs. COVTrack reuses its RoI feature heads and association logic for these external boxes.

The integration keeps requested ULGF reference annotations separate from predicted detections and tracking IDs. Requested generation identities are not used as tracking ground truth. The custom vocabulary includes ten RUOD classes with fish at class ID 4.

Earlier generation diagnostic

The 30 September diagnostic additionally produced five-frame and 48-frame translation compositions at 256 × 256 resolution using a source-derived fish keyframe, reviewed mask and supplied background. This established a separate, controllable composition workflow that complements the later three-model integration exercise.

Training and performance boundaries

CountGD++ and COVTrack successfully performed inference with existing checkpoints. The completed integration confirms data exchange and output generation. Future model-specific evaluation can add CountGD++ counting MAE/RMSE and detection precision/recall, alongside COVTrack TETA, HOTA, MOTA and IDF1 metrics; these scores are not established by the current run.

The integration also creates a practical basis for improving image quality and testing temporal consistency on longer sequences. Stable exported track identifiers are a promising interface-level result, while visual quality and identity accuracy require separate assessment.

Five frame execution results

The user read detection_sequence.json and tracking_sequence.json from the retry output. Their check required the same sequence identifier, matching frame indices and five frames in each file. The printed result is reproduced below.

Frame index	Detections	Tracks	Predicted IDs
0	1	1	0
1	1	1	0
2	1	1	0
3	2	2	0 and 1
4	1	1	0

The five-frame test yielded six detection instances and six corresponding tracking records. Predicted ID 0 appeared in every frame, demonstrating continuity of the exported identifier over the sequence. A second predicted ID appeared in frame 3, showing that the pipeline can represent multiple detections within one frame. Ground-truth review can determine the interpretation of this additional detection.

Execution recovery

The team successfully resolved Colab-specific runtime integration issues. Configuring MPLBACKEND=Agg enabled CountGD++ execution, and installing the required COVTrack import dependencies (lap, BDD100K, TETA and filelock) enabled progress through model initialisation.

The COVTrack input contract was then corrected to provide frame_id both at the top level and within img_info. After this targeted adjustment, the tracking stage successfully processed the generated sequence and produced the reported five-frame outputs.

Completion status

All three model stages successfully completed on the same generated sequence through resumed execution attempts, culminating in verified detection and tracking records. The final tracking stage was invoked directly after integration fixes. The earlier launcher status file reflects a superseded attempt rather than the final successful stage outputs; a clean single-command launcher run is a useful next reproducibility test.

Result location

Google Drive output folder: My Drive/connected_pipeline_results/retry_20261012_224051. Key evidence files are resolved_config.json, detection_sequence.json, per-frame detection JSON, tracking_sequence.json and per-frame tracking JSON. The original generation attempt used run_20261012_223233. Generated images remain under the configured ULGF experiment run folder in the Colab project copy unless separately backed up.

Verified runtime and final checklist

The run used a Tesla T4 with approximately 15 GB GPU memory. Colab's notebook kernel was Python 3.13.15; the model stages used separate interpreters. The system driver reported CUDA 13.0, while the model PyTorch builds used older CUDA runtimes.

Stage	Python	PyTorch	Key packages
ULGF	3.7.16	1.12.1+cu113	Diffusers 0.4.1; Transformers 4.24.0; Accelerate 0.20.3
CountGD++	3.10.22	2.1.2+cu118	torchvision 0.16.2; Transformers 4.40.2; Detectron2 0.6
COVTrack	3.8.20	1.10.0+cu113	MMCV 1.4.4; MMDetection 2.23.0; CLIP

CountGD++ and COVTrack import probes passed with CUDA available. CountGD++ GPU arithmetic passed, and COVTrack compiled GPU NMS returned the expected retained box. ULGF completed real generation. The supplied notebook transcript also records all six CountGD++ attention extension tests passing, including forward comparisons and numerical gradient checks.

Completed connection checklist

[x] Confirm compatible environments; use separate environments if their dependencies conflict.

[x] Execute the three stages in order.

[x] Run one short sequence through all three stages.

[x] Confirm COVTrack receives CountGD++ detections and produces tracking outputs.

Evidence references

1. ULGF/docs/TRAINING_RESULTS.md: October 7 training configuration, evaluation, backup and resume record.

2. ULGF/experiments/ulgf_training_free_video/BRANCH_FINAL_REPORT.md: earlier composition diagnostic and limitations.

3. integration/README.md, RUNTIME_SETUP.md and stage implementations: execution contracts and external-detection connection.

4. User-supplied October 12 Colab logs and final five-frame detection and tracking summary: actual execution evidence. Earlier CountGD++ component status documents predate this connection and are not current execution evidence.

Overall conclusion

Utraccount successfully achieved its core integration milestone: generated underwater frames flowed through CountGD++ detection into COVTrack tracking, with aligned five-frame outputs, six detections and six tracking records. The ULGF training continuation also validated checkpoint export, backup and resume. Together, these results establish a functioning and extensible research pipeline. The next phase is to strengthen model quality and quantify performance through dedicated perceptual, detection and tracking benchmarks.